# David S. Gonzalez

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## Education

**University of California - Berkeley**, BS Mechanical Engineering — GPA: 3.65/4.0

**University of California - Irvine**, MS Mechanical and Aerospace Engineering — GPA: In Progress

## Experience

**Robotics Research Intern**, Biomimetic Millisystems Laboratory — Mar 2024 - Dec 2024

- Led product viability testing for a 7-axis robot arm for use in brain stimulation
- Managed all CAD sub-assemblies, iterated with FEA, designed 40% of final components using Solidworks
- System design leading to successful implementation of 12 embedded systems, 4 more than previous prototype
- Fabricated highly dampened, low deflection t-slotted workbenches leading to 3 successful MRI experiments

**Manufacturing Research Intern**, Meta Materials Laboratory — May 2024 - Jul 2024

- Optimized resin 3D printing technology with acoustics to orient crystal growth, increased load capacity by 12%
- Performed fatigue experiments to gather data on material deformation using strain gauges
- Characterized testing parameters for acoustic field simulations in COMSOL
- Designed experimental fixtures and custom printer assemblies in Solidworks

**Propulsion Engineer**, STAR Rocket Club — Jul 2022 - Dec 2024

- Managed construction of LE2's vertically integrated, pressure-fed, liquid propulsion system
- Designed a quick disconnect system for cryogenic fill hoses, decreasing fill time by 12 seconds
- Developed nozzle contours and simulated performance using ANSYS and flow experiments
- Designed and machined multiple parts in the thruster assembly from engineering drawings

**Electronics Engineer**, SEB Rocket Club — Jul 2023 - Dec 2023

- Designed a remote actuation control system that controls ball valves within the rocket's fuel system
- Drafted circuit layouts in KiCAD for connections between PCBs and other components, modernizing hardware
- Designed and built heat-resistant casings around electrical systems, leading to multiple reliable data retrievals

## Projects

**LITS: Custom Force-Torque Sensor** — 2024

- "Tool Detection PCB" which measures deflection resistances to act as a 6-axis force torque sensor
- Each tool has a varying resistance when connected which runs the proper tool script

**CG Envelope for the Embraer 195-E2** — 2024

- Wrote a Python algorithm to calculate the CG Envelope of Embraer and Boeing airliners for varying weight distributions

**KIWI Bot: Omnidirectional Robot** — 2023

- Prototyped an omnidirectional robot with a tilt-degree controller and object recognition using OpenCV

**LYRA: Multiuse Robot Arm** — 2024

- Prototyped a cycloidal actuated robot arm with an interchangeable toolhead system, fully 3D printed

## Skills

**Developer Tools:** Solidworks, AutoDESK Fusion, Siemens NX, OnShape, ANSYS, Solidworks FEA

**CS/EE:** Python, Matlab, LaTex, Altium Designer, KiCAD, Simulink, LabVIEW, Arduino, Raspberry Pi

**Other:** Mills and Lathes, CNC (3,5 axis) , Drafting and GD&T, DAQ, 3D Printing, Soldering